

The Journey Navigation Flag: Turning Diabetes Encounters into Action

Stormont Vail Health wants to move beyond static visit records and understand patient journeys: how patients enter the system, what happens after diagnosis, and where care gaps appear. The challenge emphasizes long gaps between diagnosis and future appointments, social barriers, transportation issues, and different journey types.

Our team focused on Type 2 diabetes because diabetes care requires ongoing monitoring, labs, medication management, and follow-up. Instead of treating each visit as separate, we combined encounters, patients, diagnosis, departments, providers, social determinants, and census data into one journey-level view.

Our first finding is that many diabetes journeys show follow-up gaps. Among 22,155 Type 2 diabetes journeys, 61% had no timely follow-up observed within 4 months. The largest journey group was gap over 4 months, followed by one observed visit. Even among patients who did return for follow-up, the median first follow-up was 91 days. This means delayed follow-up is not a rare exception; it is one of the dominant patterns in the diabetes care journey. To make this more actionable for annual care planning, we also evaluated each diabetes patient separately for each year they appeared in the data. We defined a regular diabetes follow-up pattern as at least 4 diabetes-related encounters in a year, roughly one encounter every 4 months. Across 50,070 annual diabetes care records, only about one-quarter of diabetes care records showed regular yearly follow-up. Most patients appeared to have fewer diabetes-related touchpoints than our regular monitoring benchmark.

Our second finding is that care gaps were concentrated, not random. One-observed-visit journeys and gap-over-4-months journeys had the highest follow-up gap rates. More connected pathways, such as complex multi-touch journeys, were much more likely to show regular follow-up. This means SVH should not treat every diabetes patient the same way. Patients with long gaps or only one observed visit should be prioritized first for reconnection support.

To turn this analysis into action, we propose a Journey Navigation Flag. The flag asks: “Is this patient likely to fall below a regular diabetes follow-up pattern this year?” The improved model uses early information such as first diabetes-related encounter context, prior-year diabetes history, MyChart status, patient characteristics, and social-risk indicators. The best model reached ROC AUC = 0.799 and average precision = 0.917. More importantly, flagging the top 30% of patients most likely to need support captured about 38.2% of care-gap cases, with 94.9% precision among those flagged.

The flag should be used as a support tool, not a clinical judgment. If a patient is flagged, SVH could act within seven days by confirming follow-up, checking transportation barriers, helping with MyChart access, connecting patients to food, housing, financial, or utility support, or routing complex cases to a nurse navigator.

There are limitations. Some journeys may start before the data window or continue after it. Some follow-up may happen outside SVH. Social determinant data was not collected equally for every patient, so missing social-risk information should not be treated as no social need. SVH should use the Journey Navigation Flag to prioritize diabetes outreach, improve continuity of care, and move from reacting to care gaps to prevent them.